

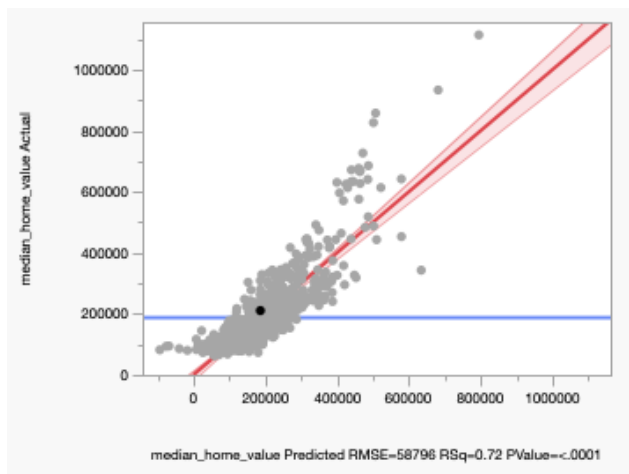
Factors Influencing the Median Home Value Across U.S. Metropolitan Areas

ILRST 2130 Project Component 3

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In this component, a full multiple linear regression model is fit to examine the relationship between median home value and the selected economic, demographic, and housing market predictors across U.S. metropolitan areas. The response variable is median home value, and the predictors include median household income, median age, total population, unemployment rate, percent renters, percent of residents with a bachelor's degree or higher, vacancy rate, and the metropolitan area code. The purpose of this analysis is to determine whether these variables collectively explain variation in housing prices and to assess which predictors remain statistically significant when considered together in a multiple regression framework.

Full Multiple Regression Model



RSquare	0.724653
RSquare Adj	0.722284
Root Mean Square Error	58796.03
Mean of Response	187550.3
Observations (or Sum Wgts)	939

Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	-667995.5	32587.11	-20.50	<.0001*
median_household_income	6.6987214	0.220818	30.34	<.0001*
median_age	4335.3567	532.9564	8.13	<.0001*
total_population	-0.00019	0.001918	-0.10	0.9212
cbsa_code	0.3274474	0.169181	1.94	0.0532
unemployment_rate	717533.21	109546.8	6.55	<.0001*
percent_renters	526428.78	43891.83	11.99	<.0001*
percent_bachelors_plus	157637.76	29775.54	5.29	<.0001*
vacancy_rate	276726.16	29188.8	9.48	<.0001*

Figure 1. Multiple regression model results for median home value.

Note. The figure displays the summary of fit, parameter estimates, and actual versus predicted values for the full multiple regression model using all predictors.

In Figure 1, a multiple linear regression model was fitted using median home value as the response variable and the seven economic, demographic, and housing predictors as explanatory variables. The model explains a substantial portion of the variation in home values across metropolitan areas.

The model has an **R² of 0.725** and an **adjusted R² of 0.722**, indicating that approximately 72% of the variation in median home values is explained by the predictors in the model.

Using the parameter estimates from JMP, the fitted regression equation is:

$$\begin{aligned}
\text{Predicted Median Home Value} = & \\
& - 667,995.5 + 6.6987(\text{Median Household Income}) \\
& + 4,335.36(\text{Median Age}) \\
& - 0.00019(\text{Total Population}) \\
& + 71,753.21(\text{Unemployment Rate}) \\
& + 52,648.78(\text{Percent Renters}) \\
& + 157,637.76(\text{Percent Bachelor's Plus})\$ \\
& + 276,726.16(\text{Vacancy Rate})
\end{aligned}$$

Global F-Test

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	8	8.4611e+12	1.058e+12	305.9443
Error	930	3.215e+12	3.457e+9	Prob > F
C. Total	938	1.1676e+13		<.0001*

Figure 2. ANOVA table for the full multiple regression model.

The global F-test evaluates whether at least one predictor has a linear relationship with median home value. From the ANOVA table, the F value is **305.94** and **p < 0.0001**. Because the p-value is far below 0.05, the global F-test is statistically significant. This provides strong evidence that at least one of the predictors is linearly associated with median home value. Therefore, the overall regression model is statistically significant.

Significance of Individual Predictors

Based on the t-tests in the parameter estimates table:

- Significant predictors ($p < 0.05$)
 - Median household income ($p < 0.0001$)
 - Median age ($p < 0.0001$)
 - Unemployment rate ($p < 0.0001$)
 - Percent renters ($p < 0.0001$)
 - Percent bachelor's degree or higher ($p < 0.0001$)
 - Vacancy rate ($p < 0.0001$)
- Not significant
 - Total population ($p = 0.921$)

This indicates that, after controlling for the other predictors, total population does not have a statistically significant independent association with median home value. All other predictors appear to contribute meaningfully to the model.

Residuals vs Fitted Values and Need for Transformation

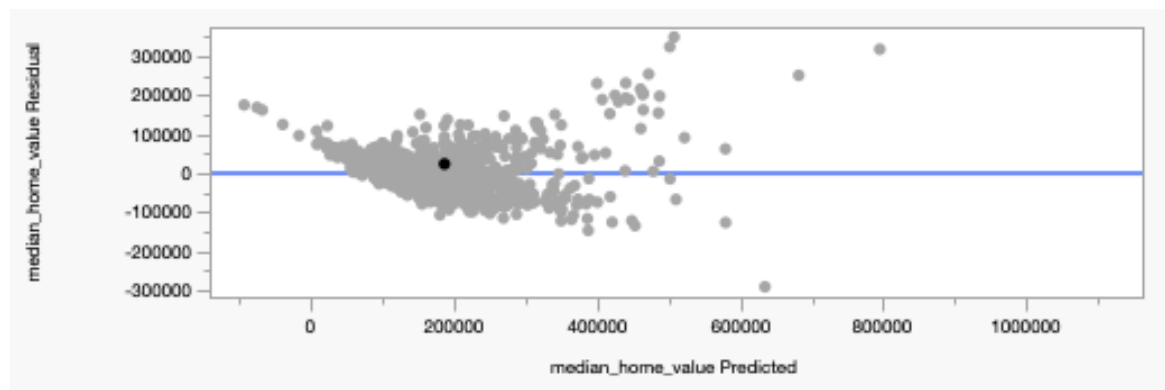


Figure 3. *Residuals versus fitted values for the original multiple regression model.*

The residuals-versus-fitted plot for the original model shows a clear fan-shaped pattern, where the spread of the residuals increases as the fitted median home values increase. This indicates heteroscedasticity, which violates a key assumption of linear regression. The residuals are more clustered at lower fitted values but become more spread out at higher predicted home values. This pattern suggests that a transformation of the response variable may be needed to stabilize the variance.

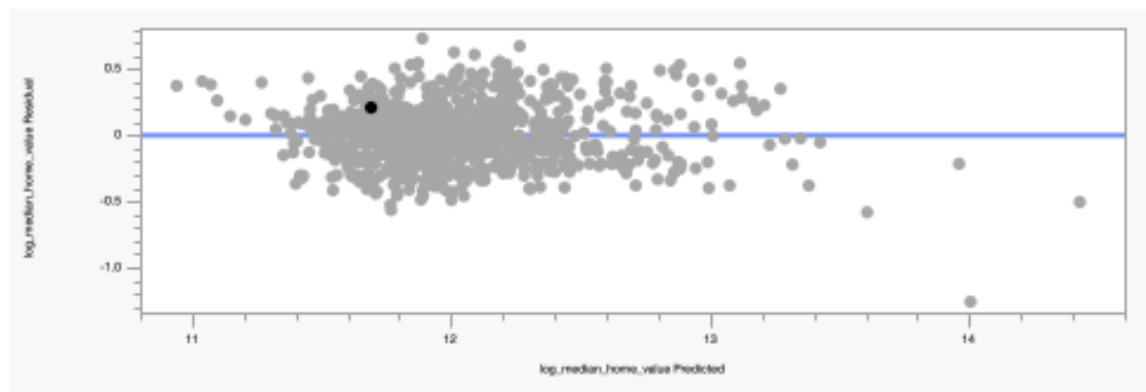


Figure 4. *Residuals versus fitted values for the log-transformed regression model.*

To address the heteroscedasticity observed in the original model, a log transformation was applied to the median home value. In the transformed model, the residuals appear more evenly distributed around zero with a more consistent spread across the range of fitted values, indicating improved variance stability. The model fit also improved slightly, with the R^2 increasing from approximately 0.72 to 0.75. Overall, the log-transformed model better satisfies the assumptions of linear regression and provides a more appropriate representation of the relationship between the predictors and median home value.